# Probability and Statistics 1B

Fergus Munro

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## 1 Probability Models and Random Variables

**Theorem.** Let $X$ be a continuous random variable with pdf $f_X(x)$ and $\mathbb{P}(a < X < b) = 1$. Suppose $g : (a, b) \to \mathbb{R}$ is continuous, strictly increasing and differentiable.

Then $Y = g(X)$ is a continuous random variable with PDF

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y))\frac{dx}{dy} & g(a) < y < g(b) \\ 0 & y \leq g(a) \text{ or } y \geq g(b) \end{cases}.$$

**Proof.** Consider $\mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y))$. Then

$$\begin{aligned}\mathbb{P}(x \leq g^{-1}(y)) &= \int_{-\infty}^{g^{-1}(y)} f_X(x)dx \\ &= \int_{a}^{g^{-1}(y)} f_X(x)dx \\ &= \int_{a}^{y} f_X(g^{-1}(y))\frac{dx}{dy}dy.\end{aligned}$$

Hence the pdf of $Y$ is

$$f_Y(y) = f_X(g^{-1}(y))\frac{dx}{dy}.$$

□

# 2 Important Families of Continuous Random Variables

## 2.1 Uniform Distribution

## 2.2 The Normal Distribution

**Definition.** The random variable $X$ has the **normal distribution** with mean $\mu$ and variance $\sigma^2$, denoted $X \sim N(\mu, \sigma^2)$ if it has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}.$$

**Definition.** The random variable $Z$ is said to be **standard normal** if it follows a $N(0, 1)$ distribution.

The pdf of the standard normal is denoted by $\phi(z)$, and the cdf denoted by $\Phi(z)$.

**Theorem.**

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} = \sqrt{(2\pi)}.$$

**Proof.** We must check that the integral of a normal distribution from $-\infty$ to $\infty$ is 1. Let

$$c = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx.$$

Then

$$\begin{aligned} c^2 &= \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dxdy. \end{aligned}$$

Now let $x = r\cos\theta$, $y = r\sin\theta$. Then $dxdy = rdrd\theta$ and

$$\begin{aligned} c^2 &= \int_0^{2\pi} \int_0^{\infty} re^{-\frac{r^2}{2}} drd\theta \\ &= \int_0^{2\pi} \left[-e^{-\frac{r^2}{2}}\right]_0^{\infty} d\theta \\ &= \int_0^{2\pi} 1 d\theta \\ &= 2\pi. \end{aligned}$$

Thus $c = \sqrt{2\pi}$. □

Then this implies that

$$\int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{\sigma^2}} dx = \int_{-\infty}^{\infty} \sqrt{2\sigma^2} e^{-\frac{u^2}{2}} du = \sqrt{2\pi\sigma^2}.$$

Then, adding the reciprocal of this to the normal pdf guarantees it will always integrate to 1.

The standard normal, $Z \sim N(0,1)$, has pdf $\phi$ and cdf $\Phi$.

**Lemma.**

- $\phi(-z) = \phi(z)$
- $\Phi(-z) = 1 - \Phi(z)$.

**Proof.** By symetry. □

**Proposition.** Suppose $Z \sim N(0,1)$ and $Y = \mu + \sigma Z$. Then $Y \sim N(\mu, \sigma^2)$.

**Theorem.** Let $0 < p < 1$ and $S_n \sim Binom(n,p)$. Then for any $a \in \mathbb{R}$

$$\mathbb{P}\left(\frac{S_n - np}{\sqrt{np(1-p)}} \leq a\right) \to \Phi(a) \text{ as } n \to \infty.$$

## 2.3 Exponential Distribution

**Definition.** The random variable $X$ has an exponential distribtion with

rate parameter $\lambda > 0$, denoted Exp($\lambda$), if it has pdf

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

and cdf

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

It has expectation

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

and variance

$$Var[x] = \frac{1}{\lambda^2}.$$

**Proposition.** Let $X$ be a random variable with $X \sim Exp(\lambda)$. Then $X$ is memoryless, meaning

$$\mathbb{P}(X > t + s | X > t) = \mathbb{P}(X > s).$$

**Definition.** The hazard rate at time $t$ of survial time distribution is

$$h(t) = \lim_{\delta t \to 0} \frac{1}{\delta t} \mathbb{P}(X \in (t, t + \delta t) | X > t).$$

This can be viewed as the instantaneous rate of survival...
If $X$ is continuous, then

$$\mathbb{P}(X \in (t, t + \delta t)) = f_x(t).$$

Thus,

$$h(t) = \frac{f_X(t)}{1 - F_X(t)}.$$

For an exponential distribution, $h(t) = \lambda$. Indeed, the exponential distribution is derived from the properties of constant hazard rate and memorylessness.

**Definition.** The random variable $X$ has a Weibull distribution with parameters $\lambda$ and $\beta$, denoted $Weib(\lambda, \beta)$, if it has pdf

$$f_X(x) = \begin{cases} \lambda \beta x^{\beta - 1} e^{-\lambda x^\beta} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The value of $\beta$ shapes the hazard rate function $h(t)$. For $\beta = 1, h(t)$ is constant, and $Weib(\lambda, 1) = Exp(\lambda)$. For $\beta > 1$, $h(t)$ increase with $t$, old items are more likely to fail than new ones, for $\beta < 1, h(t)$ decreases with $t$, old items are less likely to fail than new ones.

## 2.4 The Gamma Distribution

**Definition.** The Gamma function is defined for $t > 0$ as

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx.$$

**Lemma.**

$$\begin{aligned} \Gamma(t+1) &= \int_0^\infty x^t e{-}x dx \\ &= t\Gamma(t). \\ &= t(t-1)\Gamma(t-1). \quad \text{and so on} \end{aligned}$$

**Proof.** Use integration by parts. □

We can calculate that $\Gamma(1) = 1$, and hence we can show that for positive integers

$$\Gamma(t) = (t-1)!.$$

**Definition.** The random variable $X$ has a Gamma distribution with parameters $\lambda$ and $k$ ($\lambda > 0, k > 0$), denoted $Gamma(\lambda, k)$, if it has PDF

$$f_X(x) = \begin{cases} \frac{1}{\Gamma(k)} \lambda^k x^{k-1} e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

The Gamma Distribution has $\mathbb{E}[X] = \frac{k}{\lambda}$ and $Var[X] = \frac{k}{\lambda^2}$
The $\Gamma(\lambda, 1)$ distribution is equal to the $Exp(\lambda)$ distribution.

**Proposition.** If $X \sim \Gamma(\lambda, k)$ and $Y = cX$, then

$$Y \sim \Gamma(\frac{\lambda}{c}, k).$$

**Proposition.** If $X_1, ..., X_k$ are independent variables where $X_i \sim Exp(\lambda) \forall i$, then

$$\sum_{i=1}^k X_i \sim \Gamma(\lambda, k).$$

Hence for integers $k_1$ and $k_2$, if $Y_1 \sim \Gamma(\lambda, k_1)$ and $Y_2 \sim \Gamma(\lambda, k_2)$ then

$$Y_1 + Y_2 \sim \Gamma(\lambda, k_1 + k_2).$$

**Proposition.**

- If $X \sim N(0, 1)$, then $X^2 \sim \Gamma(\frac{1}{2}, \frac{1}{2})$.

- If $X_1, ..., X_k$ are independent $N(0,1)$ random variables, then

$$\sum_{i=1}^{k} X_i^2 \sim \Gamma(\frac{1}{2}, \frac{k}{2}).$$

The second distribution is also known as the $\chi^2$ distribution on $k$ degrees of freedom.

# 3 Generating a Random Sample from a given Distribution

## 3.1 Methods for Generating $Unif(0,1)$ Random Variables

If, for some large $m$, we can randomly arrange the numbers in the set $\{1, ..., m-1\}$ into a sequence $X_1, X_2, ...$, then we can treat $\frac{X_1}{m}, \frac{X_2}{m}, ...$ as samples from an independent $Unif(0,1)$ distribution.

**Example.** A way to do this is the following algorithm:

Set

$$m = 2^N \quad \text{for a large value of } N$$

Choose

$$c \quad \text{a prime number}$$

$$a \quad \text{such that } a-1 \text{ is divisible by c.}$$

Set $X_1$ to be any value in $\{0, ..., m-1\}$
Then proceed iteratively setting

$$X_{n+1} = (aX_n + c) \pmod{n}.$$

## 3.2 The Inverse CDF Method for Simulation

To turn a random sample from a $Unif(0,1)$ RV into a sample from a RV with CDF $g(x)$.

Consider the case where $g(x)$ is continuous and strictly increasing $g : [a,b] \to [0,1]$ with $g(a) = 0$ and $g(b) = 1$.

Then the function $g^{-1} : [0,1] \to [a,b]$ is well defined.

**Lemma.** Suppose the continuous random variable $X$ has CDF $G(x)$, and $Y = G(X)$.
Then $Y \sim Unif(0,1)$.

**Theorem.** Let $G(x)$ is a continuous, strictly increasing function from $[a,b] \to$

$[0,1]$ with $G(a) = 0$ and $G(b) = 1$. The cases $a = -\infty$ and $b = \infty$ are also allowed.

Let $U \sim Unif(0,1)$ and set $X = G^{-1}(U)$.

Then $X$ is a random variable with CDF $G(x)$.

# 4 Modelling Data and Parameter Estimation

## 4.1 Estimation by the Method of Moments

We start with the simple case of modelling the distribution of a single random variable.

Here, we have to choose a suitable distribution and estimate parameters of this distribution.

The method of moments tells us to fit the properties of chosen distribution to the properties of our sample, for example fitting $\mathbb{E}[X]$ and $Var[X]$.

**Example.** The distance between two fish in a tank is believed to follow a $Beta(a,b)$ distribution. To estimate $a$ and $b$, we need two pieces of information, for example, $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

With the observed values $x_1, ..., x_n$, we can calculate $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$, using these to find $a$ and $b$.

**Definition.** For $k \geq 0$, the $k$th moment of a random variable $X$ is

$$\mu_k = \mathbb{E}[X^k] \quad \text{(if this exists)}.$$

We assume $X_1, ..., X_n$ are i.i.d. (independent, identically distributed) and follow a distribution with unknown parameters $\theta_1, ..., \theta_n$. We can then estimate $\mu_1, ..., \mu_k$ from the observed data and use these to calculate $\theta_1, ..., \theta_n$.

**Definition.** Method of Moments Estimates of $\theta_1, ..., \theta_p$ are the solutions to

$$\frac{1}{n}\sum_{i=1}^{n} x_i^k = \mu_k = \mu_k(\theta_1, ..., \theta_p) \text{ for k } = 1, ..., p$$

where $x_1, ..., x_n$ denote the observed values of $X_1, ..., X_n$

# 5 Estimates and Estimators

**Definition.** An **estimate** is a real number computed from the data. An **estimate** of the paramter $\theta$ based on observations $x_1, ..., x_n$ can be written as

$$\hat{\theta} = h(x_1, ..., x_n)$$

for an appropriate function $h$.

**Definition.** An **estimator** is a random variable, and is a function of the random variables $X_1, ..., X_n$ that comprise the data. If the **estimate** of $\theta$ based on observations $x_1, ..., x_n$ is $\hat{\theta} = h(x_1, ..., x_n)$, the **estimator** $T$ is the random variable

$$T = h(X_1, ..., X_n).$$

## 5.1 Maximum Likelihook Estimation

Suppose the random variables $X_1, ..., X_n$ follow a certain type of distribution but the parameters of said distribution are unknown.

**Definition.** Let $X_1, ..., X_n$ be i.i.d discrete random variables with PMF

$$P(X_i = x) = p_X(\theta, x_i) \quad x_i \in \Omega \quad \text{for } i = 1, ..., n$$

where $\theta$ denotes the parameter or a vector of parameters.

Syppose values $X_i = x_i, i = 1, ..., n$ are observed, and let $bfx = (x_1, ..., x_n)$.

The likelihood function for these discrete random variables is

$$L(\theta, x) = \prod_{i=1}^{n} p_X(\theta, x_i).$$

**Definition.** Let $X_1, ..., X_n$ be i.i.d continuous random variables with PDF $f_X(\theta, x_i)$ where $\theta$ denotes the parameter or a vector of parameters.

Syppose values $X_i = x_i, i = 1, ..., n$ are observed, and let $bfx = (x_1, ..., x_n)$.

The likelihood function for these discrete random variables is

$$L(\theta, x) = \prod_{i=1}^{n} p_X(\theta, x_i).$$

**Definition.** In both the continuous and discrete cases, the log-likelihood function is

$$\mathcal{L}(\theta, x) = log\{L(\theta, x)\}.$$

**Definition.** The **Maximum Likelihood Estimate** (MLE) of a parameter of vector of parameters, $\theta$, is the value of $\theta$ that maximises the liklihood function $L(\theta, \mathbf{x})$ for the observed data.

## 5.2 Assesment of Goodness of Fit

**Definition.** Suppose we observe i.i.d. RVs $X_1, ..., X_n$. Denote the ordered

values of the observed data $x_1, ..., x_n$ by

$$x_{(1)} \leq ... \leq x(n).$$

Let $F_X$ be the CDF of a distribution proposed for the RVs $X_i$. Then, a **Quantile-Quantile plot** is a graph of

$$x_{(i)} \text{ against } F_X^{-1}\left(\frac{i}{n+1}\right).$$

If the RVs $X_1, ..., X_n$ do indeed come from the distribution with CDF $F_X(x)$, points in the Q-Q plot should lie close to a straight line through the origin with gradient 1.

# 6 Joint Distributions

## 6.1 Joint PMFs and Joint PDFs

**Definition.** We say random variables $X$ and $Y$ are jointly continuous if there eists a function $f_{X,Y} : \mathbb{R}^2 \to [0, \infty]$ such that for any region $A \subseteq \mathbb{R}^2$

$$\mathbb{P}((X, Y) \in A) = \int\int_A f_{X,Y}(x, y)\ dy\ dx.$$

The function $f_{X,Y}$ is called the joint PDF of $X$ and $Y$.

**Definition.** The joint CDF of random variables $X$ and $Y$ is the function $F_{X,Y} : \mathbb{R}^2 \to [0, 1]$ defined by

$$F_{X,Y} = \mathbb{P}(X \leq x, Y \leq y).$$

If $X$ and $Y$ are jointly continuous random variables with joint PDF $f_{X,Y}$ then

$$F_{X,Y}(x, y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u, v)\ dv\ du.$$

## 6.2 Marginal PDFs

**Proposition.** If $X$ and $Y$ have joint PDF $f_{X,Y}$, then $X$ and $Y$ have PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy.$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dx.$$

**Proof.**

$$\begin{aligned}\mathbb{P}(a \leq X \leq b) &= \mathbb{P}(a \leq x \leq b, -\infty < Y < \infty) \\ &= \int_a^b \int_{-\infty}^{\infty} f_{X,Y}(x,y)dydx \\ &= \int_a^b f_X(x)dx.\end{aligned}$$

Thus $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(y)dy$ is the PDF of $X$. □

**Definition.** Let $X$ and $Y$ be continuous random variables with joint PDF $f_{X,Y}$ and marginal PDFs $f_X$ and $f_Y$. For a value $x$ with $f_X(x) > 0$, the **conditional pdf** of $Y$ given $X = x$, written $f_{Y|X}(y|X = x)$ is

$$f_{Y|X}(y|x) = f_{Y|X}(y|X = x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \quad \text{for } y \in \mathbb{R}$$